MLOps Assignment 01

Heart Disease UCI Dataset - End-to-End ML System

Machine Learning Operations (MLOps) - AIMLCZG523

Student: _____

GitHub Repository: _____

1. Objective and Problem Statement

The objective is to build a cloud-ready machine learningclassifier that predicts heart disease risk from patienthealth information.

The project uses the Heart Disease UCI Cleveland datasetand converts the original target 0-4 into a binary targetwhere 0 means no disease and values greater than 0 meandisease.

The solution is an educational MLOps demonstration and isnot a medical diagnostic system.

- Reproducible preprocessing
- Two classifiers
- MLflow tracking
- CI/CD
- Docker
- Kubernetes
- Monitoring

2. Data Acquisition and EDA

The project downloads the Cleveland subset from UCI,converts missing-value markers to missing values, andcreates EDA visualizations.

The final submission should use the actual charts producedby the student's run.

- Class balance
- Numeric histograms
- Correlation heatmap
- Age versus maximum heart rate

3. Feature Engineering and Model Development

Numerical features use median imputation and standardscaling. Categorical/discrete features use most-frequentimputation and one-hot encoding.

Logistic Regression is the simple baseline. Random Forestprovides a nonlinear alternative. Random Forest is tunedwith five-fold stratified cross-validation using ROC-AUC.

- 80/20 stratified split
- Random state 42
- Accuracy, precision, recall, ROC-AUC

4. Experimental Results

Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	0.8852	0.8387	0.9286	0.9665
Random Forest Tuned	0.8852	0.8387	0.9286	0.9545
Random Forest	0.8689	0.8125	0.9286	0.9426

Use the actual output fromreports/figures/model_comparison.csv and explain themodel-selection decision.

5. MLflow Experiment Tracking and Reproducibility

MLflow stores local experiment runs, parameters, metrics,and plots. The complete preprocessing and model pipelineis saved using joblib.

requirements.txt and Docker provide repeatable environments.

- MLflow experiment
- ROC and confusion matrix artifacts
- Saved model
- Saved metadata

6. Testing and CI/CD

Pytest checks preprocessing and API behavior. GitHubActions runs linting, data download, tests, training,Docker build, and artifact upload.

The workflow is designed to fail on lint or test errors.

• pytest -q

• flake8

• training

• Docker build

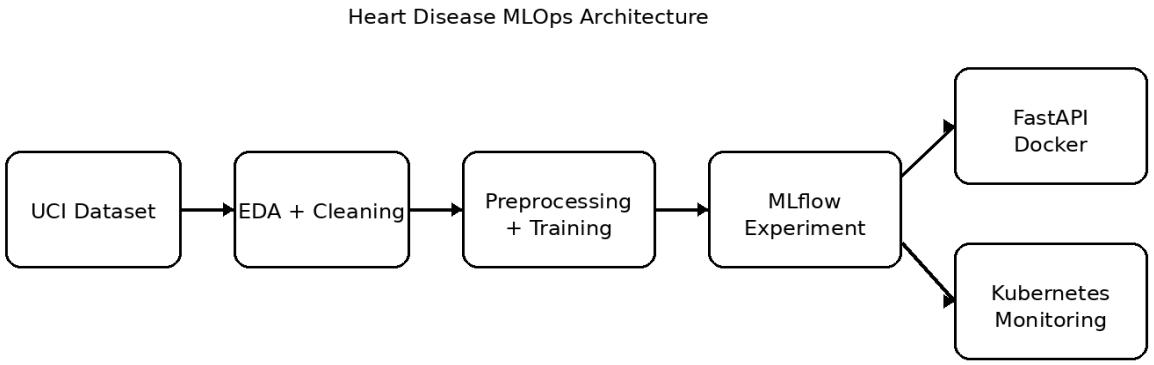
7. Containerization and API

FastAPI provides /predict, /health, and /metrics. Dockerpackages the service and trained model in an isolatedenvironment.

- Swagger /docs
- Health endpoint
- Prediction endpoint
- Metrics endpoint

8. Kubernetes Deployment and Monitoring

The Kubernetes manifest creates two replicas and aLoadBalancer Service with readiness and liveness probes.Application logs are written to standard output andPrometheus-compatible metrics are exposed at /metrics.



9. Conclusion, Limitations and Submission Checklist

The project covers the requested MLOps lifecycle from dataacquisition to deployment and monitoring.

The system is intended for academic learning. Realclinical use would require extensive additionalvalidation, security, governance, and medical oversight.

- Actual metrics inserted
- Screenshots captured
- GitHub Actions green
- Docker verified
- Kubernetes verified
- Video recorded
- Repository URL added